



ADAMJEE GOVT. SCIENCE COLLEGE

PRELIMINARY EXAMINATION-2026

MATHEMATICS-II (Science General Group)

Time allowed: 20 minutes

Section – A

Maximum Marks: 20

Multiple Choice Questions (MCQs)

Note: All notations are used in their usual meaning. The usage of scientific calculator is allowed.

Q. 1. Choose the correct answer for each of the given options.

- i). To differentiate the function $5xy$ with respect to x twice, Maple command is:
 * $> \text{diff}(5, x, y, 2x)$ * $> \text{diff}(5xy * x^2)$ * $> \text{diff}(5xy, x^2)$ * $> \text{diff}(5 * x * y, x^2)$
- ii). To create a 2×3 matrix with 1 2 3 and 4 5 6 in first and second rows respectively, Maple command is:
 * $> \text{Matrix}([(1, 2, 3), (4, 5, 6)])$; * $> \text{Matrix}([(1, 2, 3), [4, 5, 6]])$;
 * $> \text{Matrix}((1, 2, 3), (4, 5, 6))$; * $> \text{Matrix}([1, 2, 3], [4, 5, 6])$;
- iii). If $f : [-1, 4] \rightarrow \mathbb{R}$ is given by $f(x) = x^2$ then $f(-3) =$
 * ± 9 * -9 * 9 * does not exist
- iv). $\lim_{x \rightarrow 0} (1 - x)^{\frac{1}{x}} =$
 * e^3 * $e^{-\frac{1}{2}}$ * e * e^{-1}
- v). If $y = \sqrt{\tan x - y}$, then $\frac{dy}{dx} =$
 * $\frac{\tan x}{2y+1}$ * $-\frac{\operatorname{cosec}^2 x}{2y-1}$ * $\frac{\sec^2 x}{2y-1}$ * $\frac{\sec^2 x}{2y+1}$
- vi). f is the function with rule $f(x) = \ln 2x$, ($x > 0$), if g is the inverse of f , then $g'(x) =:$
 * $\frac{2}{x}$ * $\frac{1}{2x}$ * $\frac{2}{e^x}$ * $\frac{e^x}{2}$
- vii). The n th derivative of $2e^x$ is:
 * $2ne^x$ * $2e^{nx}$ * $2e^x$ * $\frac{e^x}{2}$
- viii). If S is the distance covered by a car in time t , then $\frac{d^2S}{dt^2}$ will be:
 * velocity * torque * acceleration * momentum
- ix). Let $\vec{a} = 2\hat{i} + 3\hat{j} + 5\hat{k}$ then $\frac{d}{dx} \vec{a} =$
 * 10 * $\sqrt{38}$ * 0 * 38
- x). $\int 2e^k dx =:$
 * $2e^k$ * $2xe^k$ * $\frac{2}{k+1} e^{k+1}$ * $2(k+1)e^{k+1}$
- xi). $\int \frac{dx}{a^2+x^2} =:$
 * $\tan^{-1} \frac{x}{a} + c$ * $\frac{1}{a} \tan^{-1} \frac{x}{a} + c$ * $\frac{1}{a} \sec^{-1} \frac{x}{a} + c$ * $\sin^{-1} \frac{x}{a} + c$
- xii). Two lines are said to be parallel, if and only if their slopes are:
 * equal * reciprocal to each other
 * additive inverse to each other * negative reciprocal to each other
- xiii). If the line coincides with x - axis then its equation is:
 * $y = b$ * $x = a$ * $y = 0$ * $x = 0$
- xiv). The circle $x^2 + y^2 + 20x - 8y + 16 = 0$
 * touches x - axis * touches y - axis * cuts y - axis * passes through origin
- xv). The length of tangent to the circle $x^2 + y^2 + 2y - 1 = 0$ from $(5, 2)$ is:
 * $\sqrt{24}$ units * $\sqrt{33}$ units * $\sqrt{32}$ units * $\sqrt{34}$ units
- xvi). The latus rectum of ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is:
 * $\frac{5}{32}$ * $\frac{32}{5}$ * $\frac{50}{4}$ * $\frac{4}{50}$
- xvii). The eccentricity of rectangular hyperbola is:
 * 1 * 2 * $\sqrt{3}$ * $\sqrt{2}$
- xviii). The order and degree of the differential equation is $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^3}$ is:
 * $2, 2$ * $3, 2$ * $2, 3$ * $2, 1$
- xix). The solution of differential equation $\frac{dy}{dx} + y^2 = 0$ is:
 * $y = ce^n$ * $y = \frac{1}{x} + c$ * $y = \frac{-x^3}{3} + c$ * $y = \frac{x^2}{3} + c$
- xx). In Simpson $\frac{1}{3}$ method, the number of subintervals is the multiple:
 * 4 * 1 * 2 * 3



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Section – B

(Short-Answer Questions – Marks 40)

Q. 2. Attempt TEN part questions from this section. All questions carry equal marks.

- (i). Given that $f(x) = x + 2$ and $g(x) = 3x - 2$ are two functions then show that $(gof)^{-1} = f^{-1}og^{-1}$.
- (ii). Find the derivative of function $f(x) = \sin \sqrt{x}$ OR $f(x) = \frac{1}{\sqrt{x}}$ either by ab-initio method or definition.
- (iii). Check the continuity of the function $\vec{G}(t) = (t + 8)\hat{i} + \frac{6}{t-8}\hat{j} + \ln(t + 8)\hat{k}$ at $t = 0$.
- (iv). Compute the definite integrals $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cdot \cos^2 x \, dx$ by using basic properties.
- (v). Attempt either part (a) or part (b)
 - (a) Evaluate $\int \tan^3 x \sec^3 x \, dx$
 - (b) Find the root of $3x - \cos x - 1 = 0$ by Newton's Raphson Method, correct up to 4 decimal places
- (vi). Attempt either part (a) or part (b)
 - (a) Find the area, above the x -axis under the curve $y = \frac{1}{\sqrt{4-x^2}}$ between the ordinates $x = \frac{1}{2}$, $x = \frac{\sqrt{3}}{2}$.
 - (b) Obtain the differential equation by eliminating arbitrary constant from the relation $y = A \cos x + B \sin x$.
- (vii). Attempt either part (a) or part (b)
 - (a) $A(2, 7)$ and $B(-4, -8)$ are coordinates of the line segment $\overline{AB}$. There are two point that trisect the segment $\overline{AB}$. Find the points of trisection.
 - (b) Find the equation of the parabola whose focus is $(3, 4)$ and directrix is the line $x + y - 1 = 0$.
- (viii). Attempt either part (a) or part (b)
 - (a) If $A(2, 5)$, $B(3, 7)$ and $C(0, 8)$ are the vertices of a triangle then find the equation of altitude through vertex B .
 - (b) Find the equation of the circle which passes through the two points $(b, 0)$ and $(-b, 0)$, whose radius is a units.
- (ix). Attempt either part (a) or part (b)
 - (a) Find the equation of the hyperbola with center at origin, eccentricity is 2 and vertex at $(0, 4)$.
 - (b) Find the equation of tangent to $x^2 + y^2 = 36$ with the slope $\sqrt{3}$
- (x). Attempt either part (a) or part (b)
 - (a) Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for the function $f(x) = \sin(x + y)$.
 - (b) Obtain the first three terms of the Maclaurin's series for xe^{-x} .

SECTION – C (Detailed-Answer Questions – Marks 40)

Note: Attempt FIVE questions. All questions carry equal marks.

Q. 3. Evaluate any TWO of the following integrates.

- (a). $\int \cos \sqrt{x} \, dx$
- (b). $\int \frac{\sec^2 x}{(1+\tan x)(2+\tan x)} \, dx$
- (c). $\int_0^2 \frac{dx}{\sqrt{16-x^2}}$

Q. 4. Attempt either part (a) or part (b)

(a). Evaluate limit of the functions.

$$\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$$

$$\lim_{x \rightarrow \infty} \left(1 - \frac{pq}{x}\right)^{\frac{x}{p}}$$

(b). Differentiate the following.

$$y - xy - \sin y = 0$$

$$y = \sqrt{\frac{1+\cos x}{1-\cos x}}$$

$$y = \sin^{-1}(\cos x) + \cos^{-1}(\sin x)$$

Q. 5. Attempt either part (a) or part (b)

(a). Find a right-angled triangle of maximum area with a hypotenuse of length h .

(b). Evaluate the integral $\int_0^{\frac{\pi}{4}} \sin x \, dx$ with 6 intervals by using Simpson's $\frac{3}{8}$ rule.

Q. 6. Attempt either part (a) or part (b)

(a). Find the equation of a line through the intersection of the lines $2x + 7y - 8 = 0$ and $3x + 2y + 5 = 0$ and making an angle of 45° with the line $2x + 3y - 7 = 0$.

(b). Find the equation of ellipse with center at $(0, 1)$ and major axis is parallel to y -axis. Also, it passes through $(2, 1)$ and $(0, 4)$.

Q. 7. Attempt either part (a) or part (b)

(a). A resistor of 5 ohms and a capacitor of 0.02 farads are connected with 10 volts battery. Assume that initially charge on capacitor is 5 (coulombs). Find the charge and current in the circuit at any time.

(b). Solve the differential equation $\frac{dy}{dx} = \frac{x-2y+2}{2x+y-1}$